

Model-Based Parametric Mapping Reconstruction

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15.1 Introduction

The aim of parameter mapping is to determine quantitative physical and/or biophysical parameters ($\mathbf{v}$) that provide diagnostic information. This chapter will focus on the estimation of MR relaxation times (T_1 , T_2 , T_2^* ...), but the methods discussed here can in principle also be applied to other approaches for quantitative parameter estimation such as model-based fat–water estimation (see Chapter 14).

Relaxation times are determined by quantum mechanical properties of nuclear spins. Even before the first MR images were acquired, it was shown that pathological changes of tissue can lead to changes of these relaxation times [1,2]. Nowadays, mapping of relaxation times is used in a wide range of various clinical applications. T_1 can be used to detect and characterize tumors or to assess changes in the brain due to epilepsy or multiple sclerosis. Changes of T_1 times in the myocardium can be an indication for the presence of fibrosis or fat in the heart muscle. Inflammatory processes can increase the amount of water retained in tissue and thus lead to changes of T_2 . T_2^* mapping can be used to detect iron depositions in tissue. In addition to the parameters that are of diagnostic interest, often additional parameters such as the spin density ρ also need to be estimated. Therefore, $\mathbf{v} = [\mathbf{v}_1, \dots, \mathbf{v}_M]$ where each $\mathbf{v}_i$ represents a spatial map of one parameter such as T_1 or ρ .

The main advantage of using mapping approaches for clinical applications is that they provide quantitative parameters that are easy to compare between different scans, scanners, patients, and institutions. The standard qualitative images (e.g., T_1 or T_2 weighed images), on the other hand, strongly depend on hardware and sequence specific parameters, which can make a comparison very challenging. As already mentioned, relaxation times are not biophysical parameters such as blood flow but are determined by the nuclear spins. Therefore, there can be a dependency of $\mathbf{v}$ on the field strength of the MR scanner. Nevertheless, clinical scanners operate at well-defined field strengths of 1.5T or 3T, making this less of a problem.

The main challenges of parameter mapping are long acquisition times and more complex signal processing compared to qualitative MRI. Section 15.2 introduces some basic aspects of MR data acquisition for parameter mapping, but the main focus of this chapter will be on image reconstruction. For more information on data acquisition, please refer, e.g., to [3].

In order to be able to obtain parameters $\mathbf{v}_i$ from acquired k-space data $\mathbf{s}$, the data acquisition has to be sensitive to changes in $\mathbf{v}_i$, i.e., different values for $\mathbf{v}_i$ have to lead to different k-space data $\mathbf{s}$. This is commonly achieved by using preparation pulses (e.g., inversion, saturation or T_2 -sensitive pulses) that prepare the nuclear spin system for the subsequent data acquisition. Another option is to vary

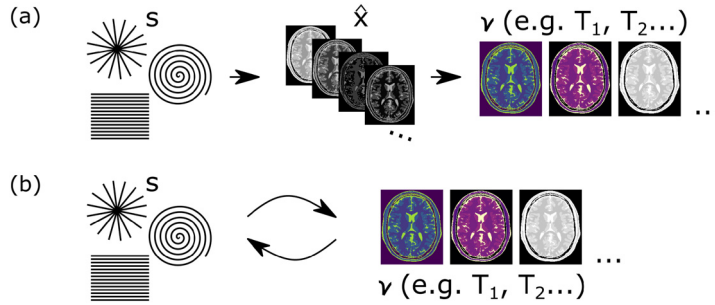


FIGURE 15.1 Parameter mapping

(a) For image-based parameter mapping, a set of qualitative images $\hat{x}$ are reconstructed, and in a second step the quantitative parameters v are estimated, which can represent T_1 , T_2 or other parameters that are unknowns in the applied model q such as the flip angle. (b) For reconstruction-based mapping, v is obtained iteratively from the acquired k-space data s . This ensures that the obtained v are consistent with s .

MR sequence parameters (e.g., flip angle (α), echo time (T_E) or repetition time (T_R)) during data acquisition. The effect of the MR sequence on the image signal x given a certain set of parameters v can be described by a signal model q . The acquired k-space data is therefore given by

$$s = E q(v) \quad \text{with} \quad q : v \mapsto q(v), \quad (15.1)$$

and, where the encoding operator E describes the entire MR acquisition process including coil sensitivity weighting, Fourier encoding and k-space sampling (see Chapter 2).

Parameter mapping, i.e., the estimation of v given s , can be carried out in two different ways (Fig. 15.1). The first approach, which we will refer to here as *image-based mapping*, separates Eq. (15.1) into two subproblems: image reconstruction and parameter estimation. First, images $\hat{x}$ are reconstructed from s . Each image shows a different image contrast due to the underlying differences of the parameter of interest v , e.g., to estimate T_1 , $\hat{x}$ can be images at different time points after an inversion pulse or for T_2 mapping, $\hat{x}$ can be images acquired after different T_2 -sensitive preparation pulses. In an independent second step v is estimated from the reconstructed images $\hat{x}$ using an appropriate signal model q . The second class of techniques, solves Eq. (15.1) directly (*reconstruction-based mapping*). These two methods are discussed in Sections 15.3 and 15.4 in more detail.

15.2 MR mapping sequences

In order to be able to estimate v from the acquired MR data, the data acquisition has to be sensitive to v , i.e., different combinations of v need to result in different MR signals. Due to the flexibility of MRI, there is a wide range of approaches to how the MR data acquisition can be made sensitive to T_1 , $T_2, \dots$ or a combination of these parameters.

Fig. 15.2 shows simulated signal curves (i.e., longitudinal magnetization) of three different species (i.e., $T_1 - T_2$ combinations) for a gradient-echo sequence. For constant sequence parameters (i.e., T_R ,

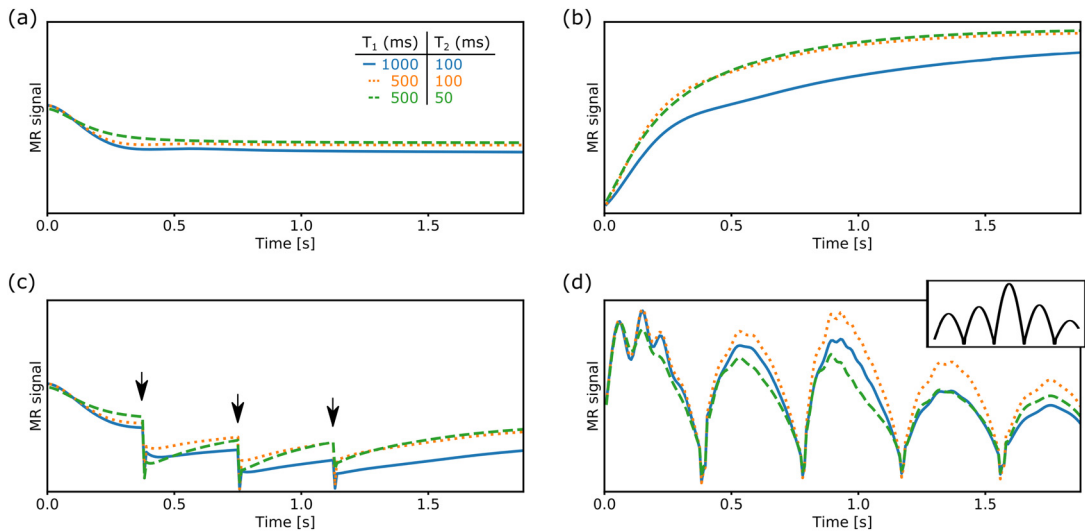


FIGURE 15.2 Signal curves for different MR sequences

Signal curves were simulated for three different species (i.e., $T_1 - T_2$ combinations). (a) Without any pre-pulses or sequence parameter modulation, the species yield very similar signals. (b) An inversion pulse prior to the data acquisition makes the sequence dependent on T_1 . (c) T_2 -sensitive preparation pulses (black arrows) lead to T_2 dependency. (d) Magnetic resonance fingerprinting uses, e.g., a varying flip-angle pattern (small figure insert) to achieve both T_1 and T_2 dependency. All simulations were carried out using an extended phase graph algorithm assuming an unspoiled gradient-echo sequence.

T_E and α), all species yield very similar signal curves and cannot be distinguished (Fig. 15.2a). For T_1 mapping, commonly an inversion pre-pulse or saturation pre-pulse is applied and the signal is then acquired during T_1 relaxation (Fig. 15.2b). Another option is to use different flip angles (variable flip-angle approach) in order to achieve signal variation as a function of T_1 . Fig. 15.2c shows T_2 dependency achieved by multiple T_2 -sensitive preparation pulses, e.g., a combination of $90^\circ - 180^\circ - 180^\circ - 90^\circ$ pulses, which lead to T_2 weighting of transversal magnetization which is then flipped back to the longitudinal direction and can be measured. T_2 mapping can also be carried out using spin-echo acquisitions at different echo times and similarly multi-echo gradient-echo acquisitions can be used for T_2^* mapping.

The examples discussed so far aim at achieving MR signal dependency mainly on a single parameter. A multi-echo spin-echo sequence with appropriately set T_R depends only on T_2 and a spoiled low-flip angle gradient echo sequence is highly T_1 dependent. The advantage of this approach is that simple signal models (e.g., mono-exponential functions) can be used to describe the MR signal as a function of $\mathbf{v}$, which allows for robust parameter mapping. Nevertheless, this requires different acquisitions for different parameters and hence can make a comprehensive assessment of multiple mapping parameters very time consuming.

A wide range of approaches has been proposed that overcome this challenge by obtaining multiple parameters $\mathbf{v}_i$ from a single scan (see, e.g., [4–11]). These methods provide different diagnostic parameters such as T_1 or T_2 that are perfectly spatially coregistered allowing for a direct voxel-wise

comparison of both parameters. In order to be able to estimate multiple ν_i , the MR sequence has to be sensitive to all these parameters, which is often achieved by varying multiple sequence parameters. An example is shown in Fig. 15.2d for Magnetic Resonance Fingerprinting (MRF) [6]. A varying flip angle (small figure insert in Fig. 15.2d) leads to signal dependency on both T_1 and T_2 . Commonly, this is also combined with an inversion preparation pulse to achieve a strong T_1 effect. The main challenge here is that such signal variations cannot be described by a single exponential model, but the evolution of the MR signal has to be calculated using more complex models such as Bloch equations or Extended Phase Graph (EPG) approaches [12]. These methods ensure that the entire spin system is traced over time, and the recorded signal contains the appropriate contributions from gradient echoes, spin echoes, and stimulated echoes.

In order to estimate $\mathbf{v}$, it is assumed that the sequence parameters are well known. This is in general true for the sequence timings (e.g., T_R , T_E). Parameters, such as the flip angle α on the other hand, can deviate strongly from the theoretical value. This of course means that the model used to describe the MR data is inaccurate, leading to errors in the parameter estimation. In order to minimize these errors, the sequence design can be optimized (e.g., using longer RF excitation pulses to achieve a more accurate α), or these sequence parameters have to be included as unknowns in the signal model and also estimated during the mapping. Therefore, many of the multi-parametric methods already mentioned do not just provide several diagnostic parameters but also additional maps of, e.g., B_0 or B_1 fields or other model parameters.

The flexibility of MR offers a wide range of possibilities to design sequences for parameter mapping. It is therefore crucial that both the MR sequence and the MR signal model are adapted to a given mapping problem. In addition, the limitations of the MR sequence (e.g., inaccurate α) need to be well understood and need to be taken into considerations during mapping.

15.3 Image-based mapping

Image-based parameter mapping can be divided into three steps: sampling of the magnetization over time (Fig. 15.3a), reconstruction of qualitative images with different contrast weighting (Fig. 15.3b), and prediction of the parameter of interest (Fig. 15.3c). Image reconstruction and parameter estimation are usually independent of each other, with parameter estimation implemented as a post-processing step, performed voxel-wise based on the qualitative images.

The data acquisition is chosen such that $\mathbf{s}$ is sensitive to the parameter to be resolved, with preferably low sensitivities to other undesired biophysical parameters. Independent of the used method, a series of qualitative images $\hat{\mathbf{x}}_i$ has to be sampled to resolve the evolution of the magnetization ($\mathbf{M}(t)$), e.g., T_1 recovery after an inversion pulse. This is achieved by sampling of $\mathbf{M}(t)$ at multiple predefined time points τ_i leading to multiple sets of k-space data $\mathbf{s}_i$ (Fig. 15.3a).

From $\mathbf{s}_i$, multiple contrast-weighted qualitative images $\hat{\mathbf{x}}_i$ are reconstructed. Ideally, each $\mathbf{s}_i$ is obtained within an infinitesimally short time compared to the temporal changes of $\mathbf{M}(t)$. In practice this is often not possible and hence the changes of $\mathbf{M}(t)$ over the acquired $\mathbf{s}_i$ can also lead to artifacts (e.g., image blurring). Furthermore, restrictions due to clinically feasible scan time can also mean that not a fully sampled k-space can be acquired for each τ_i , but undersampling is necessary. Commonly, uniform Cartesian sampling is used for $\mathbf{s}_i$, but for certain applications non-Cartesian (e.g., radial or spiral trajectories) or nonuniform Cartesian k-space sampling is preferred because it can lead to incoherent

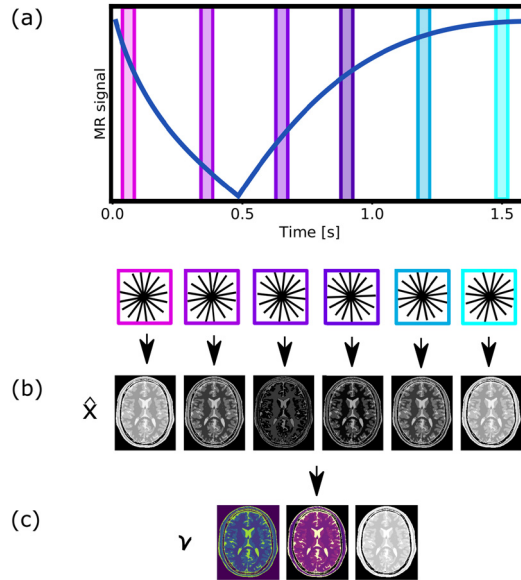


FIGURE 15.3 Overview of image-based parameter mapping

(a) After magnetization preparation, $\mathbf{s}$ is acquired at various time points using, e.g., a radial acquisition scheme. The blue (dark gray in print version) curve shows the temporal evolution of the absolute value of the magnetization for an arbitrary T_1 time. (b) Images $\hat{\mathbf{x}}_i$ are then reconstructed from $\mathbf{s}_i$. (c) From the signal evolution in each voxel of $\hat{\mathbf{x}}_i$, parameters $\mathbf{v}$ are estimated by, e.g., carrying out a fit of the data to a signal model q . In this example, $\mathbf{v}$ consists of three parametric maps showing T_1 , the equilibrium magnetization M_0 , and the flip angle α . Only T_1 is clinically relevant, but the other two are required to ensure the model describes the signal evolution well. There are, of course, other approaches, that provide multiple diagnostic parameters as discussed in Section 15.2.

undersampling artifacts. Incoherence of the undersampling artifacts not just within each $\hat{\mathbf{x}}_i$ but also between different $\hat{\mathbf{x}}_i$ is advantageous because these artifacts propagate less into the parameter maps $\mathbf{v}$. This requires a different k-space sampling pattern for each $\hat{\mathbf{x}}_i$. For imaging affected by physiological motion (e.g., breathing or heartbeat), triggering and gating, or motion correction (see Chapter 13) is needed, which can make it challenging to cover the entire temporal evolution of $\mathbf{M}(t)$.

The qualitative images $\hat{\mathbf{x}}_i$ can be reconstructed using multiple reconstruction methods that affect the image quality of $\hat{\mathbf{x}}_i$ and therefore $\mathbf{v}$. Fig. 15.4 compares various image reconstruction schemes for undersampled $\mathbf{s}_i$ and the effect on the obtained T_1 map. The simplest approach is to use FFT or NUFFT (see Chapter 4) to reconstruct $\hat{\mathbf{x}}_i$ (Fig. 15.4a). Nevertheless, because $\mathbf{s}_i$ is undersampled, undersampling artifacts are present that also affect the T_1 map. An improvement of $\hat{\mathbf{x}}_i$ can be achieved by using parallel imaging (see Chapter 6), which also increases the quality of the T_1 map (Fig. 15.4b). Both approaches treat each $\hat{\mathbf{x}}_i$ separately and do not utilize the fact that each image shows the same anatomy, just with a different image contrast. Image reconstruction can thus be further improved by applying regularization along the temporal dimension of $\hat{\mathbf{x}}_i$. This can be carried out independently of the underlying physical

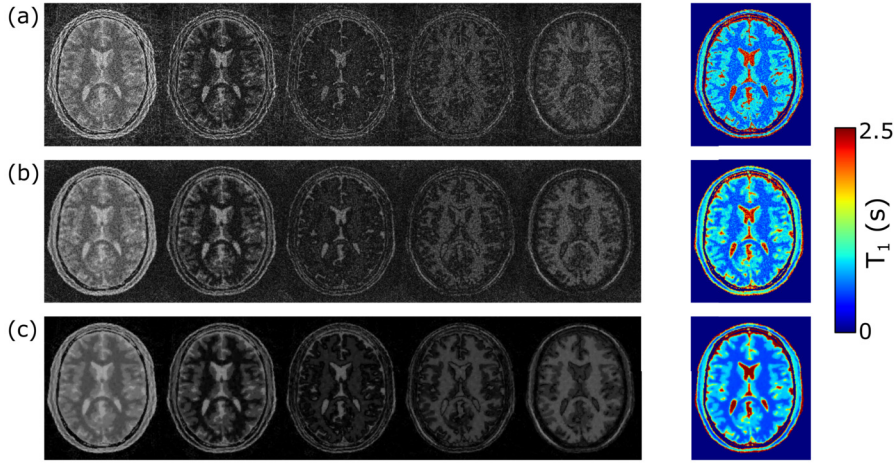


FIGURE 15.4 Image reconstruction approaches for direct T_1 mapping

2D radial data acquisition after an inversion pulse was simulated. Images at multiple time points after the inversion pulse were reconstructed from highly undersampled data using (a) nonuniform FFT, (b) iterative non-Cartesian SENSE and (c) iterative reconstruction with TV-based regularization in space and over time. T_1 maps were determined using a voxel-wise fit with a Levenberg–Marquardt approach.

model q utilizing, e.g., TV-based regularization (Fig. 15.4c) or enforcing sparsity along the temporal dimension. Temporal acceleration techniques such as kt-SENSE or kt-PCA can also be used [13]. Low-rank based-reconstruction can also be applied (see Chapter 9) which assumes that groups of voxels follow similar temporal behavior without having to explicitly specify the temporal model [14,15,10,16]. Another option is to already utilize q during image reconstruction in order to calculate a dictionary-based sparsifying transform [17,18].

As shown in Fig. 15.4, the image quality of $\hat{\mathbf{x}}_i$ affects the quality of the obtained parameter map. Adding regularization can strongly improve $\hat{\mathbf{x}}_i$, especially for highly undersampled $\mathbf{s}_i$. Nevertheless, regularization needs to be carefully chosen for the reconstruction task at hand, and the regularization strength also needs to be optimized. If the regularization is not well chosen, regularization artifacts (e.g., staircasing artifacts for regularization using total variation [19]) can appear in $\hat{\mathbf{x}}_i$ and can thus also affect the final $\mathbf{v}$, which needs to be taken into consideration when interpreting $\mathbf{v}$. For more details on regularized image reconstruction, please also see Chapter 2.

Based on $\hat{\mathbf{x}}_i$, $\mathbf{v}$ is predicted. This is done voxel-wise, e.g., by L2-norm minimization of $\hat{\mathbf{x}}_i$ and q of the specific acquisition. For example, for T_1 mapping, an inversion preparation could be used, where the magnetization recovery can be derived from the Bloch equations

$$\frac{dM_z}{dt} = \frac{M_0 - M_z}{T_1}, \quad (15.2)$$

resulting in

$$M_z(t) = M_0 + (M_z(t_0) - M_0)e^{-t/T_1}, \quad (15.3)$$

with $M_z(t_0)$ the initial magnetization at $t = 0$. Another approach is a variable flip-angle method, where at least two gradient-echo acquisitions with different flip angles are acquired. These can be described by the signal model

$$M_z(T_R, \alpha) = M_0 \frac{1 - e^{-T_R/T_1}}{1 - \cos \alpha e^{-T_R/T_1}} \sin \alpha, \quad (15.4)$$

and T_1 can be calculated based on the signal from both acquisitions.

T_2 can be estimated using a spin-echo sequence that can be described by

$$M_{xy}(t) = M_0 e^{-t/T_2}. \quad (15.5)$$

Next to these monoexponential models, the Bloch model or EPG can be used to obtain q .

Multiple relaxation parameters can be important for an accurate classification of a disease so more recent approaches employ the combination of $\mathbf{v}$ in a single acquisition (see also Section 15.2). One of these approaches is MRF. Here, data acquisition is performed using pseudorandomly chosen TR and flip angles instead of keeping these values constant during a MR scan. This leads to a signal intensity sensitive to T_1 as well as T_2 , with a specific signal evolution for each combination of these parameters, a so-called “fingerprint”.

Instead of conventional fitting, dictionary matching is performed to obtain $\mathbf{v}$. For this, a dictionary $\mathbf{D}$ is generated for this specific acquisition, with signal time curves $\mathbf{d}_i$ of a wide range of physiologically possible combinations of $\mathbf{v}$. The Bloch equations (see also Chapter 1) or EPG [12] are commonly used to calculate the dictionary entries. Usually, this step has to be performed only once for a specific MRF sequence. Eventually, $\hat{\mathbf{x}}$ is matched voxel-wise to $\mathbf{D}$ to obtain the best $\mathbf{v}$. This is done by computing the dot product between the normalized $\hat{\mathbf{x}}$ and all entries $\mathbf{d}_i$ of $\mathbf{D}$ ($\mathbf{d}_i \cdot \hat{\mathbf{x}}$), as a correlation-based metric. The best $\mathbf{v}$ is given by the dictionary entry with the highest dot product ($i_{\max}$). M_0 , proportional to the proton density, is calculated by the scaling factor between $\hat{\mathbf{x}}$ and $\mathbf{d}_{i_{\max}}$. One thing to keep in mind is that the dictionary $\mathbf{D}$ is always only a discrete representation of all possible values of $\mathbf{v}$. If $\mathbf{D}$ only coarsely resolves, e.g., T_1 , then the same will be true for the obtained map $\mathbf{v}$.

For MRF sequences that have to be synchronized with physiological motion such as the heart beat, $\mathbf{D}$ has to be calculated for each scan because the physiological motion determines the timings of the sequence and hence $\mathbf{M}(t)$ [20]. In order to minimize artifacts due to cardiac motion, the MR scan can be ECG-triggered such that data is only acquired in a specific cardiac phase. This improves the final quality of $\mathbf{v}$ but also means that the timings of the MR scan depend on the heart rate.

One of the main drawbacks of MRF is the large range of combinations of parameters. This leads to a very large dictionary and computationally demanding pattern recognition. A common approach to speed up the process is using singular value decomposition (SVD)

$$\mathbf{D} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^*. \quad (15.6)$$

Based on this, a low-rank approximation of the dictionary can be calculated

$$\mathbf{D}_k = \mathbf{D} \mathbf{V}_k, \quad (15.7)$$

with $\mathbf{V}_k$ containing the first k right singular vectors of $\mathbf{D}$ [21]. For compression in the time domain, the time evolution of the dictionary entries $\mathbf{d}$ can be projected onto the low-rank space

$$\mathbf{d}_k = \mathbf{d} \mathbf{V}_k, \quad (15.8)$$

and pattern matching can be performed in this subspace. Many other approaches have been proposed to speed up the dictionary matching and reduce its memory requirements using, for example, group matching or linear interpolation [22–25]. For more details on MRF, please refer [26] to and [27]. As a starting point, there are several open-source software packages for fast calculation of $\mathbf{D}$ available on github.^{1,2}

Machine learning can also be applied in parameter mapping at any stage of the process. For example, image reconstruction could be performed with a neuronal network to replace iterative reconstructions and thus to speed up image reconstruction (see Chapter 10). Furthermore, it could also be applied for data fitting, fast dictionary generation, or pattern recognition [28–30].

Parameter estimation as part of image postprocessing has the advantage that it is easy to implement because the parameters are obtained separately for each voxel. Furthermore, since the estimation has only to be performed once for each voxel, this method is fast, and the total time of parameter mapping depends largely on the data acquisition and the image reconstruction method used.

One of the main challenges is that artifacts in $\hat{\mathbf{x}}$ (e.g., due to coherent undersampling, mismatch of motion states, or field inhomogeneities) can impair $\mathbf{v}$ since no data consistency between $\mathbf{v}$ and $\mathbf{s}$ is included.

15.4 Reconstruction-based mapping

Undersampling of k-space speeds up data acquisition but can lead to residual artifacts in $\hat{\mathbf{x}}$. Depending on the artifacts, this can lead to quantification inaccuracies in $\mathbf{v}$. No information of the encoding model $\mathbf{E}$ is included in the model operator q , and hence it is not possible to distinguish between true signal and undersampling artifacts.

Reconstruction-based mapping tries to overcome this problem by directly solving Eq. (15.1), i.e., estimating $\mathbf{v}$ from given k-space data $\mathbf{s}$. The main advantages of this approach are that data consistency with the acquired k-space data can be ensured and regularization can be applied directly on $\mathbf{v}$ rather than the intermediate $\hat{\mathbf{x}}$. Rather than estimating $\mathbf{v}$ from image data impaired by undersampling artifacts, the quantitative parameters are directly estimated from $\mathbf{s}$. Assuming we have an image with $N \times N$ voxels and want to estimate N_v quantitative parameters from N_a quantitative images $\hat{\mathbf{x}}$, then, for image-based mapping, we need to estimate $N \times N \times N_a$ unknowns from $\mathbf{s}$. For the reconstruction-based mapping, we need to estimate $N \times N \times N_v$ unknowns and as N_v is often much smaller than N_a , this is a better determined problem. The main challenge of the model-based approach is that Eq. (15.1) describes a nonlinear problem. In the following we will describe two approaches to solve this problem. Model-based acceleration of parameter mapping (MAP) separates the problem of finding $\mathbf{v}$ into several individual steps that each on its own can be solved in a straightforward way. Model-based optimization directly solves Eq. (15.1) using a nonlinear optimization.

¹ https://github.com/chixindebaoyu/MRF_CUDA.

² <https://github.com/utcsilab/mri-sim-py.epg>.

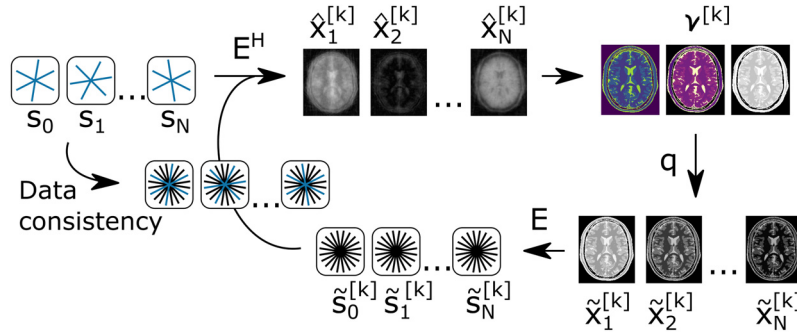


FIGURE 15.5 Overview of MAP

Starting from the acquired k-space $\mathbf{s}_i$, first image estimates $\hat{\mathbf{x}}_i^{[k=1]}$ are reconstructed and $\mathbf{v}^{[k]}$ is estimated. In each iteration k , k-space data $\tilde{\mathbf{s}}_i^{[k]}$ is calculated from $\mathbf{v}^{[k]}$ using q and $\mathbf{E}$, and data consistency is ensured. More and more of the missing k-space data is filled in at each iteration, which improves the estimation of $\mathbf{v}$. Please refer to Section 15.4.1 for further details.

15.4.1 Model-based acceleration of parameter mapping (MAP)

For image-based mapping, images $\hat{\mathbf{x}}$ are reconstructed, and $\mathbf{v}$ is determined from them. The main idea of MAP is to extend this and calculate the k-space data representing $\mathbf{v}$ using q and $\mathbf{E}$. This k-space data $\tilde{\mathbf{s}}$ is then compared to the acquired data $\mathbf{s}$ and used to update $\hat{\mathbf{x}}$. This is repeated multiple times until convergence is reached.

MAP assumes a set of k-space data $\mathbf{s}_i$, where i represents the different dynamics (e.g., different inversion times for inversion-recovery based T_1 mapping). The main requirements are that the k-space sampling leads to incoherent artifacts along the dynamic dimension (e.g., spiral or radial acquisitions turned by the golden angle for each i) and that the k-space center is well covered for each dynamic i to ensure contrast changes are captured and hence mainly high k-space frequencies are undersampled. MAP then carries out the following steps for each iteration k , which are also depicted in Fig. 15.5:

1. Reconstruct an initial estimate $\hat{\mathbf{x}}_i^{[k=1]}$ of the acquired k-space data $\mathbf{s}_i$ using $\mathbf{E}^H$.
2. Estimate $\mathbf{v}^{[k]}$.
3. Calculate the qualitative image sequence $\tilde{\mathbf{x}}_i^{[k]}$ from $\mathbf{v}^{[k]}$ using q .
4. Calculate the k-space data $\tilde{\mathbf{s}}_i^{[k]}$ representing $\tilde{\mathbf{x}}_i^{[k]}$ using $\mathbf{E}$.
5. Ensure data consistency (e.g., by substituting the acquired k-space positions in $\tilde{\mathbf{s}}_i^{[k]}$ with $\mathbf{s}_i$).
6. Repeat steps 2–5 until convergence has been reached.

In this example, the acquired radial k-space $\mathbf{s}_i$ has a higher sampling density in the center of k-space. Therefore, the initial images $\hat{\mathbf{x}}_i^{[k=1]}$ reconstructed by applying the adjoint of the encoding operator $\mathbf{E}^H$ contain very little high-resolution information. Commonly, this is compensated for using a density compensation function to ensure a balanced contribution of high- and low-k-space frequencies to the reconstructed image (see also Chapter 4). For the example shown in Fig. 15.6, no density compensation function is applied, leading to a higher contribution of low k-space frequencies to the image for the initial iterations. This also means that undersampling artifacts are attenuated allowing for an estimation

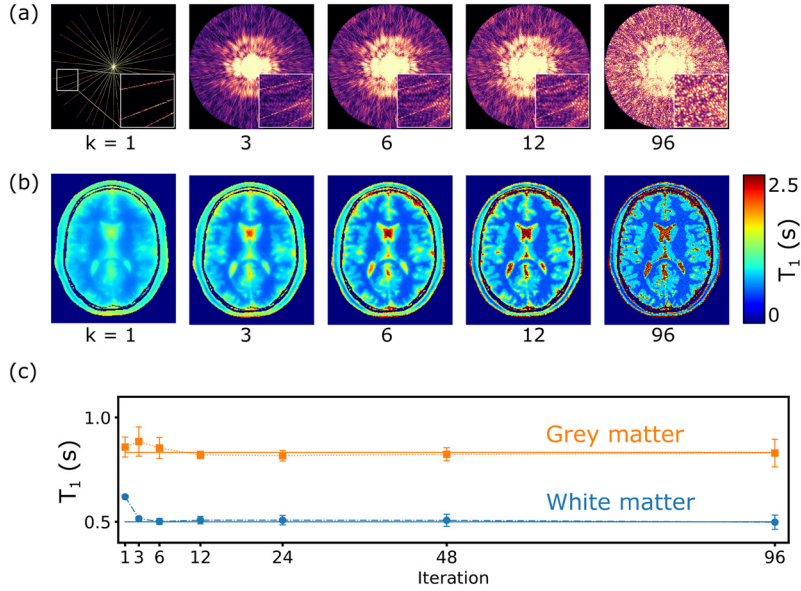


FIGURE 15.6 Convergence of MAP

(a) K-space data s_i and (b) T_1 maps at multiple numbers of iterations (k) of the MAP reconstruction. (c) Convergence of T_1 values for region-of-interests in gray and white matter to the ground truth values (dashed lines).

of $\mathbf{v}^{[k]}$ with fewer artifacts but, of course, also fewer image details. If too many artifacts are present in $\hat{\mathbf{x}}_i^{[k=1]}$, then additional low-pass filters can also be applied to s_i prior to image reconstruction. The low-pass filter strength can then be decreased in each iteration to allow for more and more high-resolution information in the images and in the parametric maps [31].

The estimation of $\mathbf{v}^{[k]}$ from $\hat{\mathbf{x}}_i^{[k=1]}$ can be carried out using fitting algorithms, dictionary matching-based approaches, or any other parameter-estimation technique discussed in Section 15.3. Despite $\hat{\mathbf{x}}_i^{[k]}$ being low-resolution images, they might still be impaired by undersampling artifacts, and therefore the method used to estimate $\mathbf{v}^{[k]}$ needs to be robust in the presence of these artifacts.

Fig. 15.6 shows $\mathbf{v}^{[k]}$ for various iteration numbers. For this reconstruction, a 2D golden-radial acquisition after a single inversion pulse was simulated. Each s_i consisted of eight radial lines at various time points after the inversion pulse. A Look-Locker model [32] was used as q and, in addition to T_1 , also the spin density and the actual flip angle was estimated as $\mathbf{v}$. The radial sampling leads to a higher density of k-space points in the center of k-space and hence the first reconstruction using $\mathbf{E}^H$ leads to low-resolution images $\hat{\mathbf{x}}_i^{[k]}$ and hence a low-resolution estimation of $\mathbf{v}^{[k=1]}$. With each iteration, more and more details are visible in $\mathbf{v}^{[k]}$ as more and more of the missing k-space data at higher k-space frequencies are filled in. Fig. 15.6c depicts T_1 values of a region-of-interest in the white and gray matter as a function of the number of iterations. For higher iteration numbers, the mean values approach the true value (dashed lines) but also the standard deviation in each region increases. Undersampling is higher for higher k-space frequencies, and hence the data consistency step has less of an effect here.

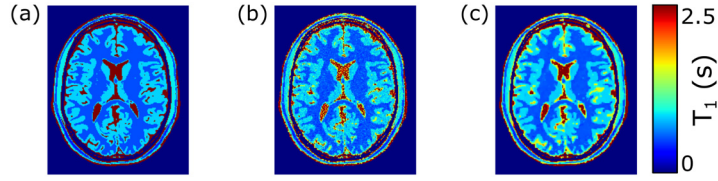


FIGURE 15.7 MAP with de-noising

(a) Ground truth T_1 map. (b) T_1 map after 96 iterations using MAP without de-noising. (c) T_1 map after 96 iterations with TV-based de-noising.

With increasing number of iterations, more of the higher k-space data get filled in but also noise starts to build up. The number of iterations has to be therefore carefully selected to ensure all image features are recovered and an accurate estimation of $\mathbf{v}$ is achieved before noise starts to dominate the data. In order to further stabilize the reconstruction and avoid the problem of noise amplification for high numbers of iterations, de-noising of $\mathbf{v}^{[k]}$ can be added as a reconstruction step. Fig. 15.7 compares the ground truth T_1 map (a) to the estimated T_1 maps after 96 iterations without de-noising (b) and with TV-based de-noising (c). With the de-noising step, the reconstruction can be further stabilized, and the problem of noise dominating the reconstruction for higher iteration numbers is avoided. Of course, this requires a suitable selection of a de-noising approach adding further parameters to the reconstruction and leading to a potential bias of the final result.

MAP has been used for T_1 and T_2 estimation and a similar approach has also been proposed for MRF [33,31].

15.4.2 Model-based optimization

As just mentioned, Eq. (15.1) describes a highly nonlinear problem. Nevertheless, a minimization of a likelihood cost function is possible to estimate the quantitative parameters $\hat{\mathbf{v}}$ from the k-space data $\mathbf{s}$

$$\hat{\mathbf{v}} = \operatorname{argmin}_{\mathbf{v}} \|\mathbf{n}(\mathbf{v}) - \mathbf{s}\|_2^2, \quad (15.9)$$

with

$$\mathbf{n} : \mathbf{v} \mapsto \mathbf{E}q(\mathbf{v}) \quad (15.10)$$

and using the L2-norm. Commonly, this problem is extended by further regularization to stabilize the solution

$$\hat{\mathbf{v}} = \operatorname{argmin}_{\mathbf{v}} \|\mathbf{n}(\mathbf{v}) - \mathbf{s}\|_2^2 + \sum_i \lambda_i R_i(\mathbf{v}) \quad . \quad (15.11)$$

Suitable regularization terms R_i have to be adapted to the requirements of each problem. Examples of proposed regularizations are:

- piecewise constant $\mathbf{v}$ by minimizing the L1-norm of TV or total generalized variation of $\mathbf{v}$ [34];
- smoothness of $\mathbf{v}$ by minimizing the L2-norm of the first-order difference matrix of $\mathbf{v}$ [35];

- smoothness of the k-space data by minimizing the L2-norm of $\nabla \mathcal{F} \mathbf{v}$, where ∇ is the spatial finite-difference operator and $\mathcal{F}$ the spatial Fourier transform [36];
- sparsity of $\mathbf{v}$ and similarity between $\mathbf{v}_i$ using wavelets as sparsifying transforms and ensuring that wavelet coefficients that are present in all $\mathbf{v}_i$ are favored [37].

Similar to the choice of regularization, the choice of the solver for (15.11) also depends strongly on $\mathbf{E}$ and q , which is still an active field of research. Olafsson et al. linearized the problem of T_2^* mapping from a multi-echo gradient-echo acquisition and used a conjugate gradient approach [35]. Sbrizzi et al. used a variable projection method [9], and other groups proposed iteratively regularized Gauss–Newton methods [36,38,34].

In order to solve this problem iteratively, the Jacobian matrix of Eq. (15.9) needs to be calculated

$$\begin{aligned}
 \frac{\partial}{\partial \mathbf{v}} \|\mathbf{n}(\mathbf{v}) - \mathbf{s}\|_2^2 &= \frac{\partial}{\partial \mathbf{v}} \|\mathbf{E}q(\mathbf{v}) - \mathbf{s}\|_2^2 = \frac{\partial}{\partial \mathbf{v}} \left((\mathbf{E}q(\mathbf{v}) - \mathbf{s})^H (\mathbf{E}q(\mathbf{v}) - \mathbf{s}) \right) \\
 &= \frac{\partial}{\partial \mathbf{v}} (\mathbf{E}q(\mathbf{v}))^H (\mathbf{E}q(\mathbf{v}) - \mathbf{s}) + (\mathbf{E}q(\mathbf{v}) - \mathbf{s})^H \frac{\partial}{\partial \mathbf{v}} (\mathbf{E}q(\mathbf{v})) \\
 &= \frac{\partial}{\partial \mathbf{v}} (\mathbf{E}q(\mathbf{v}))^H (\mathbf{E}q(\mathbf{v}) - \mathbf{s}) + \left(\frac{\partial}{\partial \mathbf{v}} (\mathbf{E}q(\mathbf{v}))^H (\mathbf{E}q(\mathbf{v}) - \mathbf{s}) \right)^H \\
 &= 2\text{Re} \left(\frac{\partial}{\partial \mathbf{v}} (\mathbf{E}q(\mathbf{v}))^H (\mathbf{E}q(\mathbf{v}) - \mathbf{s}) \right) \\
 &= 2\text{Re} \left(\left(\frac{\partial q(\mathbf{v})}{\partial \mathbf{v}} \right)^H \mathbf{E}^H (\mathbf{E}q(\mathbf{v}) - \mathbf{s}) \right),
 \end{aligned} \tag{15.12}$$

where we used $A + A^H = 2\text{Re}(A)$ with $\text{Re}(A)$ referring to taking the real part of A .

Approaches have also been proposed that make use of the Hessian using trust-region algorithms to solve Eq. (15.9) [39]. Calculating and storing the Hessian explicitly can be computationally demanding. Therefore, sparse approximations of the Hessian have been proposed instead.

The problem in Eq. (15.11) is nonconvex, and, therefore, it is also important to provide a good starting point for the minimization in order to ensure convergence to the desired solution. The convergence rate of most solvers will strongly depend on the correct scaling of the Jacobian/Hessian matrix. The quantitative parameters $\mathbf{v}$ can represent physical quantities such as relaxation times, and hence they have units (e.g., ms or s) or they can be unitless, such as the spin density. Therefore, the various partial derivatives need to be scaled appropriately [36,37]. The same considerations apply to the strength of regularization λ_i in Eq. (15.11). The regularization type and strength have to be carefully optimized for each physical quantity of $\mathbf{v}$.

Once the objective function and the Jacobian/Hessian are defined, the problem can be treated similarly to the classical image-reconstruction problems described in Chapters 2 and 3.

Fig. 15.8 shows an example of a T_2 map reconstructed from a multi-echo spin-echo sequence acquired in a human brain using a golden-ratio radial acquisition pattern [40]. Each echo is highly undersampled. In this example, $\mathbf{v}$ representing a M_0 and T_2 map, and the coil sensitivity maps were jointly estimated using a model-based optimization. L_1 -wavelet regularization was applied to $\mathbf{v}$, and the nonlinear problem was solved using an iteratively regularized Gauss–Newton method. For comparison, a standard image-based mapping approach is also shown that uses iterative non-Cartesian SENSE

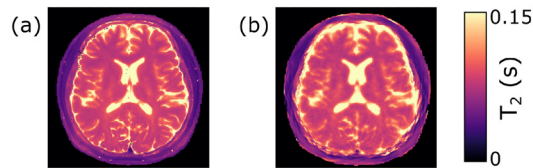


FIGURE 15.8 T_2 mapping using model-based optimization

T_2 map obtained from a multi-echo spin-echo in-vivo scan using (a) model-based optimization compared to (b) image-based mapping. This figure was reproduced from [40] using the code and data that was made publicly available with this publication.

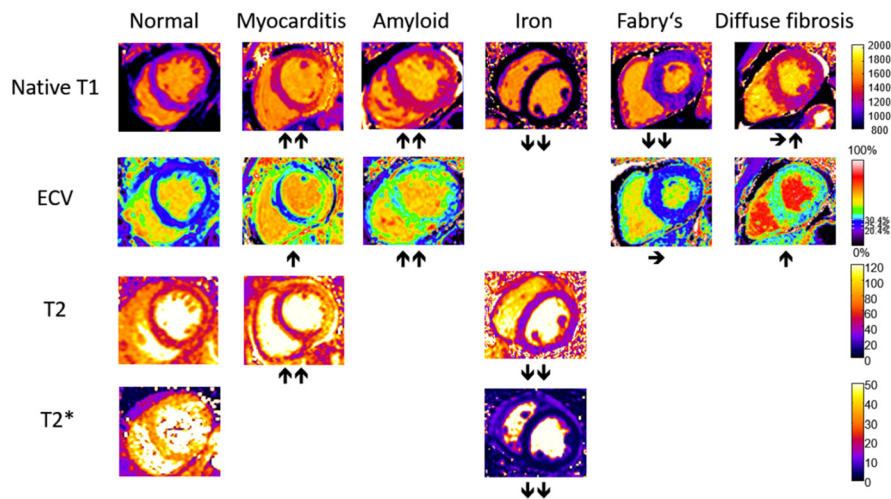


FIGURE 15.9 Parameter mapping for cardiac applications

T1, T2, T2*, and ECV (extracellular volume) mapping are useful for the diagnosis of many different myocardial diseases. The arrows denote changes to the parameter value compared to healthy myocardium. Reprinted from Messroghli et al., JCMR (2017) 19:75, <https://dx.doi.org/10.1186/s12968-017-0389-8> [42].

for the reconstruction of $\hat{\mathbf{x}}_i$, followed by a voxel-wise fit of Eq. (15.5) to the data. The model-based optimization shows a better depiction of anatomical details compared to the image-based approach.

15.5 Clinical applications

Parameter mapping can be used to improve the diagnosis and classification of pathologies in various body regions, such as the brain, heart, and abdomen [41]. Pathological processes alter the local molecular environment of tissue, implicating altered relaxation times of this tissue. Fig. 15.9 gives an overview of changes in mapping parameters for multiple pathologies of the heart. The increase of

free-water content causes prolonged T_2 times so T_2 is sensitive to edema. Edema is present in acute inflammatory diseases or acute ischemia. Furthermore, absolute changes in T_2 also make it possible to monitor inflammatory processes over time. Another application of T_2 mapping is hemorrhage, which can induce shortened T_2 times.

Iron is important for the production of hemoglobin, usually stored in the liver and spleen, but accumulation of iron can occur, damaging the organs. The superparamagnetic behavior of the iron-stored proteins leads to susceptibility-induced distortion in the local magnetic field, making T_2^* the most suitable parameter for the detection of iron storage diseases, with shortened T_2^* times. Moreover, the T_2^* time correlates with the absolute iron level in the liver, making it possible to replace biopsy for accurate diagnosis.

Infiltrative diseases are also affecting the T_1 times because of the change in tissue composition. For example, specific lipids (Fabry's disease) or insoluble proteins, such as amyloid, could be deposited in the interstitial space of the tissue, severely damaging the tissue if untreated. These changes lead to increased T_1 times (amyloidosis) or shortened T_1 times (Fabry's disease).

Native T_1 times obtained by T_1 mapping without the application of a contrast agent are sensitive to intracellular as well as extracellular changes of the tissue. However, in many diseases, especially, the extracellular space is of great importance. Using both native T_1 mapping and T_1 mapping after the administration of a contrast agent changing the T_1 times, the extracellular volume can be calculated, directly obtaining a physiologically parameter that is important in all diseases involving the remodeling of the tissue. In general, remodeling of the extracellular space is present in many diseases, thus T_1 can be used for the detection and monitoring of many pathophysiological changes, such as myocarditis, chronic diseases (e.g., multiple sclerosis, dementia) or post-transplantation changes of the kidney and liver. The T_1 value itself provides an indication of the underlying pathology and its extent. By comparing the quantitative relaxation times to healthy tissue, overall longer T_1 times could be caused by diffuse scarring fibrosis, and shortened T_1 times could be a marker for diffuse fat infiltration.

The combination of multiple parameters enables a more certain decision about the pathological state of the tissue. Therefore, methods such as MRF, combining T_1 and T_2 mapping in a short scan time could highly impact future clinical routine. As shown in Fig. 15.10, using an MRF approach, metastases in the liver caused by lung adenocarcinoma can be better visualized compared to its T_1 -weighted image, as shown in the study of Chen et al. [43].

Whereas local pathological tissue, and thus locally altered parameters, could possibly be also detected in qualitative images, the main advantage of parameter mapping could lie in the detection of diffuse diseases where no contrast to healthy tissue is present. Furthermore, because of the system- and acquisition-independent nature of T_1 and T_2 and T_2^* , parameter mapping facilitates objective interpretations. Comparability across patients or institutes is increased, making large-scale multi-center studies possible and thus enabling broader research and therefore better understanding of diseases. For this, robust and fast techniques have to be investigated to be able to obtain parameter maps during clinically feasible scan times.

15.6 Current challenges and future directions

Mapping is a very powerful technique, providing quantitative diagnostic parameters for a wide range of pathologies. The main challenge are inconsistencies between the acquired data and the applied model

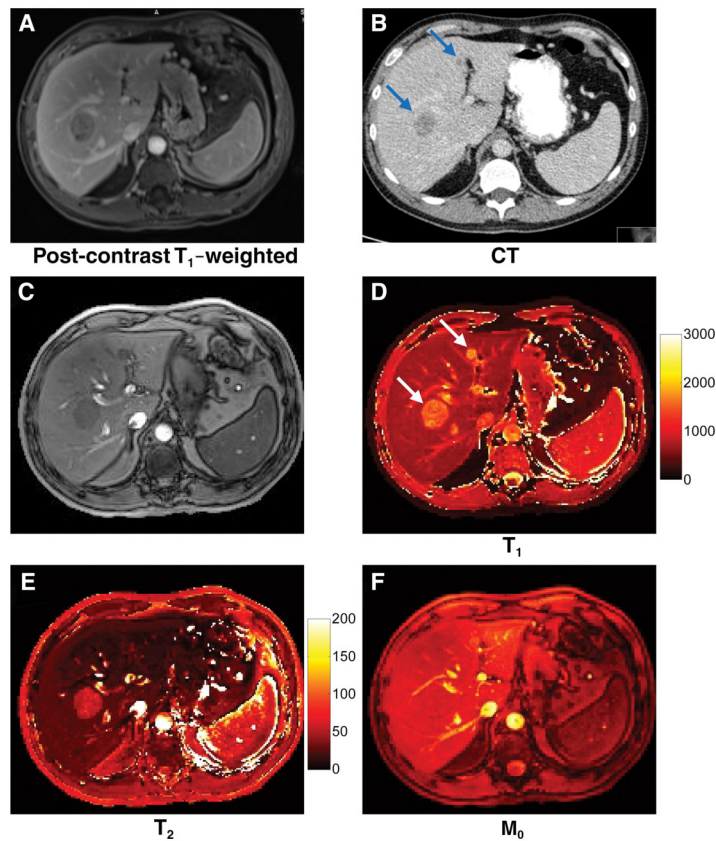


FIGURE 15.10 Parameter mapping of lung adenocarcinoma metastatic to the liver using MRF

Metastases (arrows) are more visible in parameter maps obtained by MRF (C-F) compared to a qualitative contrast-enhanced T1-weighted image (A). Reprinted with permission from Radiology Society of North America (RSNA) from Fig. 5, Chen Y, Jiang Y, Pahwa S, Ma D, Lu L, Twieg MD, Wright KL, Seiberlich N, Griswold MA, Gulani V: MR Fingerprinting for Rapid Quantitative Abdominal Imaging. *Radiology* 2016, 279:278–286 [43].

operator q leading to errors in the estimated ν . Often a simplified model (e.g., single-exponential signal recovery) is used since it provides fast mapping and is robust towards noise and other artifacts in the data. The model is commonly adapted to the data acquisition and hence different MR mapping sequences (e.g., T_1 mapping using either an inversion or a saturation pulse) lead to different model errors and hence different estimations of ν . In clinical practice, reference values for healthy and pathological tissue therefore have to be determined for each sequence, separately. Future work is required to improve the comparability of various mapping approaches.

MR relaxation times are parameters that describe the behavior of nuclear spins in a magnetic field, hence they can only be determined with MRI. Other quantitative parameters used for clinical diagnosis,

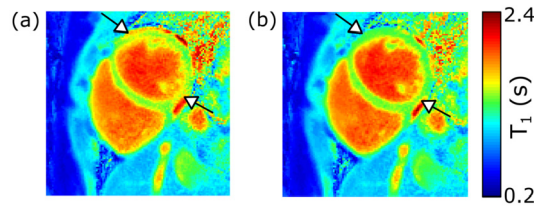


FIGURE 15.11 Fig. T_1 mapping with and without motion correction

(a) Residual cardiac motion leads to inaccurate myocardial T_1 times which would suggest the presence of a myocardial pathology (arrows). (b) By correcting for the cardiac motion, these artifacts are removed, showing a homogeneous healthy myocardium.

such as blood flow velocities, can be measured with MRI but also with ultrasound or invasively with catheters, providing reference measurements that are independent to MRI. This is not possible for mapping. MR-based reference measurements for relaxation times are available, but they are commonly very time consuming (longer than one hour) and therefore not applicable in vivo. This makes it challenging to fully evaluate the accuracy of mapping sequences and also to determine the true values for various tissue types.

One approach to minimize differences between the actual MR signal evolution and q is to make q more and more complex (e.g., by including magnetization-transfer effects) to ensure the MR physics is accurately described. Of course, this makes mapping more computationally demanding and more time consuming. In addition, it can also make it less robust in the presence of noise or residual artifacts. Recently, machine learning has also been introduced for parameter mapping. On one hand, this offers the possibility to learn q directly from the data, assuming an accurate reference measurement is available as training data. On the other hand, it can also be combined with a MR physics-based model to enable fast parameter estimation, while ensuring $\mathbf{v}$ is consistent with the acquired data [29]. Fast parameter estimation is also essential if complex models are to be used in clinical practice.

Inconsistencies between the data and the applied model can have two different effects. On the one hand, model errors can lead to a bias of the estimated $\mathbf{v}$, but still result in a good match between model and data. An example of this is given in the tutorial Section 15.8 where the effects of an incomplete inversion on the estimation of T_1 is analyzed. Although the estimated value for T_1 is inaccurate, the model fits well to the data. Such errors can only be detected if reference measurements are available. The other type of errors lead to a mismatch between model prediction and acquired data. This can also lead to errors of $\mathbf{v}$ but not necessarily. An example of this is MRF. The undersampling of each image is not taken into consideration in the signal model. Therefore, there is actually a poor match between the dictionary entries and the acquired data. Because the undersampling artifacts are incoherent compared to the signal model, an accurate estimation of $\mathbf{v}$ is still possible.

Also, physiological changes during data acquisition can lead to differences between q and the acquired data. One of the most common causes is physiological motion, such as head motion, swallowing, breathing motions, or the beating of the heart. MR scans are carried out to minimize any physiological motion, but this is not always possible. Sometimes, patients cannot hold their breath long enough during data acquisition or they experience heart rate irregularities that lead to residual motion in the data and hence artifacts in the reconstructed images. Fig. 15.11 shows an example for T_1 mapping in a

short-axis view of the heart. Respiratory motion is minimized by asking subjects to hold their breath. Nevertheless, cardiac motion is still present and can lead to errors in ν . In this case the T_1 value in some parts of the myocardium are elevated which would suggest the presence of fibrosis but is simply an artifact due to cardiac motion (Fig. 15.11) [44]. Physiological motion has to be taken into consideration to ensure accurate and reliable parameter mapping. For more details, please refer to (Chapter 13).

15.7 Summary

Mapping yields quantitative parameters that make objective diagnosis and treatment monitoring possible. Image-based mapping approaches separate image reconstruction and parameter estimation and enable fast estimation of the parameters. Parameter estimation can be carried out by data fitting, dictionary matching, or machine-learning approaches. The physical model used for parameter estimation can already be incorporated into image reconstruction to enable accurate parameter mapping even from highly undersampled data. Image reconstruction and parameter estimation can also be combined in an iterative process to further improve quantification accuracy by ensuring data consistency between the acquired raw data and the obtained parametric maps. Clinical applications for parametric mapping range from assessing inflammatory processes to detecting iron accumulation and fibrotic tissue. Nevertheless, large clinical trials are still needed to establish parametric mapping in clinical routine.

15.8 Tutorial

15.8.1 Image-based T1 mapping

15.8.1.1 Problem description

In this exercise we will carry out T_1 mapping for four tissue types (fat, white matter, gray matter and cerebrospinal fluid (CSF)), assuming data acquisition carried out with an inversion recovery sequence using the following signal model:

$$M_z(t) = M_0(1 - 2I_{eff}e^{-t/T_1}), \quad (15.13)$$

where $M_z(t)$ is the longitudinal magnetization at time point t , M_0 is the equilibrium magnetization. Ideally, an inversion pulse will lead to a full inversion of the magnetization (i.e., 180° pulse), but, due to hardware limitations and interaction of the magnetic field with the body, the inversion pulse can lead to an excitation less than 180° . This is described with the inversion efficiency $I_{eff} \in [0, 1]$ (e.g., $I_{eff} = 1$ is the ideal case and corresponds to a 180° pulse).

This exercise requires the Curve Fitting Toolbox of Matlab®.

15.8.1.2 Provided material

- **sim_sig = sig_ir(M0, T1, Ieff, t_acq, noise_level)** This function simulates data acquisition after an inversion pulse. Example: `sim_sig = sig_ir(1, 1000, 0.9, [0, 100, 200], 0.05)` simulates a signal with $M_0 = 1$, $T_1 = 1000$ ms, $I_{eff} = 0.9$, a noise level of 0.05 at 0 ms, 100 ms, and 200 ms after the inversion pulse.

- **fit_ir(sig, t_acq)** This function fits the signal model of the inversion recovery sequence to given data. Example: `fit_ir(sig, [0, 100, 200])` fits the model to a 1D signal *sig* acquired at the time points [0, 100, 200].

15.8.1.3 Questions

1. Simulate signals for fat (T_1 : 350 ms), white matter (T_1 : 500 ms), gray matter (T_1 : 833 ms) and CSF (T_1 : 2569 ms) with $M_0 = 1$, $I_{eff} = 1.0$, noise level = 0.1 at the time points $t = [0, 600, 1200, 1800, 2400, 3000]$.
2. Fit the signal model to this data and report the error between the estimate and true T_1 times for each tissue type separately.
3. Simulate the signals for fat, white, and gray matter and CSF again, but for the time points $t = [0, 1800, 2400, 3000, 3600, 4200]$. Repeat the fitting and report the errors for each tissue. Why can T_1 for white and gray matter and CSF still be well estimated, whereas T_1 for fat shows larger errors?
4. Simulate the signals for fat, white, and gray matter and CSF again, with $t = [0, 600, 1200, 1800, 2400, 3000]$ and $I_{eff} = 0.9$. This corresponds to the case when we do not achieve a full 180° inversion with the inversion pulse. Calculate the errors of T_1 . Is T_1 over- or underestimated and explain why?

15.8.2 Magnetic resonance fingerprinting

15.8.2.1 Problem description

In this exercise we will use a precalculated dictionary $\mathbf{D}$ to estimate the T_1 and T_2 of four signal curves $\hat{\mathbf{x}}$. The (T_1, T_2) combination used to calculate the entries of the dictionary are listed in $\mathbf{v}_{Dict}$. As discussed in Section 15.3, for dictionary matching we carry out the following steps:

- Calculate the dot product between $\mathbf{D}$ and $\hat{\mathbf{x}}$ for each (T_1, T_2) combination in the dictionary.
- Find the dictionary entry with the index i_{max} where this dot product is highest.
- The i_{max} entry in $\mathbf{v}_{Dict}$ then provides the estimated T_1 and T_2 values.

15.8.2.2 Provided material

- **ex_15_2_mrf_data.mat** `Dict_sig` is the precalculated dictionary ($\mathbf{D}$) of 60,000 (T_1, T_2) combinations over 1000 MR acquisition data points, `Dict_par` contains the corresponding T_1 and T_2 values ($\mathbf{v}_{Dict}$), and `Sig` are the four unknown signal curves $\hat{\mathbf{x}}$ (again, each with 1000 data points) from which we want to estimate the T_1 and T_2 values. In addition, the true T_1 and T_2 values of the unknown signals are also provided as `T1_true` and `T2_true`.

15.8.2.3 Questions

1. Carry out dictionary matching with the provided dictionary $\mathbf{D}$, report the found T_1 and T_2 values, and calculate the error compared to the true T_1 and T_2 values.
2. Carry out a singular value decomposition (SVD) of the dictionary along the 1000 data points.
3. Compress the dictionary $\mathbf{D}$ to only include the first N_k SVD components with $N_k \ll 1000$.
4. Repeat the dictionary matching for different values of N_k and report the errors of the T_1 and T_2 estimation.
5. What is the main benefit of carrying out the SVD compression?

Acknowledgment

The images of the brain simulated in this chapter to illustrate the various reconstruction techniques were created based on data from the BrainWeb: <http://www.bic.mni.mcgill.ca/brainweb/> [45]. We would like to thank Dr. Martin Uecker (University Medical Center Göttingen) for his help with reproducing Fig. 15.8.

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